

Figure 1. Study Overview: Cost-Minimization Analysis of Automated Respiratory Rate Monitoring

Literature Search

Database Search
PubMed
NICE | FDA 510(k)
PMDA | Gov Statistics

Device Identification
& Screening

9 Devices
5 Technology Categories

Bioacoustics | Bioimpedance
Piezoelectric | Radar
Capnography

Model Construction

Comparator
Manual RR Counting
vs 6 Automated Scenarios

Hospital Perspective
1-Year Time Horizon

Cost Parameters
Nurse wages | Device costs
Early-warning costs
Adverse-event costs

4 Countries
Japan | UK | Australia | USA

Analysis

**Cost-Minimization
Analysis**

Break-Even
Threshold Analysis

One-Way
Sensitivity Analysis

Multi-Way
Scenario Analysis

Cross-Country
Comparison